NTNU
Kunnskap for en bedre verden

Department of Ocean Operations and Civil Engineering

MMA4002 Design for Automated Manufacturing

Assignment 1

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1 Product concept

1.1 Concept and value proposition

The product concept chosen is family of "classy" and non-modernistic outdoor furniture with a focus on benches. Although the end user is whoever is sitting on the bench, they don't pay for it. The ones who pay, the customers, would primarily be municipalities, cities, larger companies and other similar institutions. The idea is that these larger institutions and companies care less about the price of each individual bench and more about the message the furniture conveys and how they affect the public relations and well being of the end users. There are several other mass produced designs on the market, but it seems that most of them are very modernistic. According to a recent study, most people questioned preferred traditional architecture over contemporary (Mouratidis 2026). The hypothesis of this concept is that this applies to outdoor furniture as well. In other words, the belief is that older style benches would provide more value to these institutions than the bench it self.

1.2 Visualization



(a) Model 1 - Bench (V1)



(b) Model 1 - Bench no back (V1)



(c) Model 1 - Chair (V1)



(d) Model 1 - Bench with different design.

Figure 1: Some AI-generated sketches of potential designs. Due to the modularity of the design, several different configurations of the bench, including several different armrest designs are possible.

1.3 Product family

The product family will consist of several types of outdoor furniture. At first, the idea is to focus on benches. The common platform of these products are thought to be the bench seat/back rest. These are designed so that very similar (or potentially the same) modules could be used as walls on planters and trash cans as well. This could potentially also be extended to more products in the product family like for example planters, public trash cans.

1.4 Common platform, modules and standardized interfaces

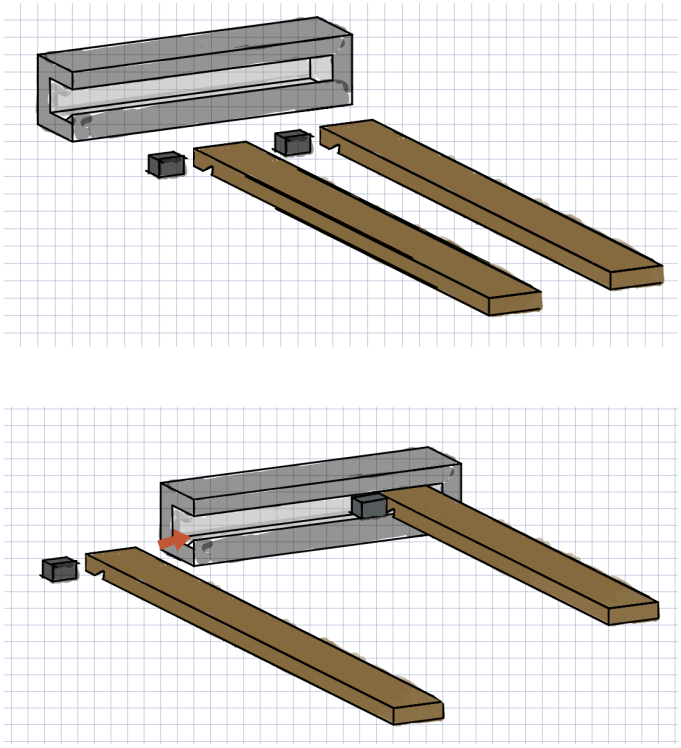
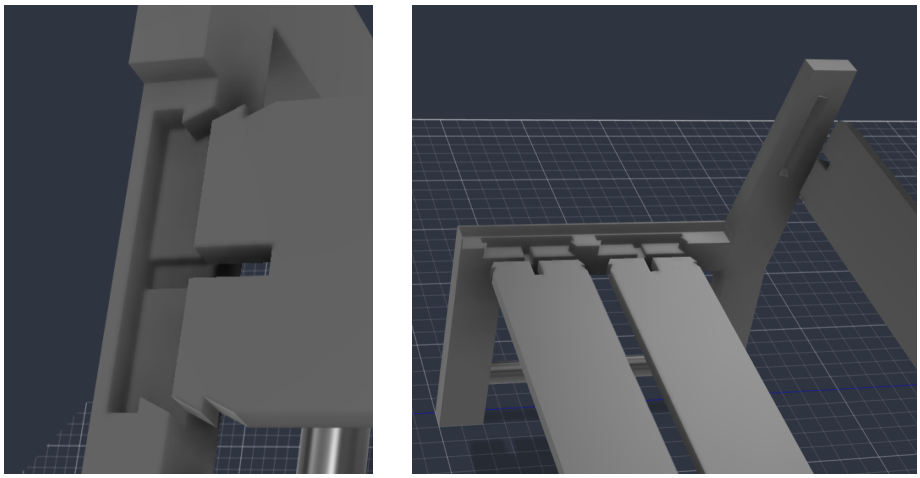


Figure 2: Hand drawn sketch of a sliding interface.

The interfaces are not decided upon yet, but several ideas has been floated. One idea is to slide planks into a track with spacers in between as seen in figure 2. The planks would then be locked in place by a component at the end of the track that presses on the planks and gives friction. This could allow for omitting screws entirely, but at the same time yield a couple of problems. Some other ideas are to use either screws, glue, press fit-mechanisms, dove tails (as seen in figure 3), friction or a combination of these.



(a) Interface V1 (b) Interface V1 - All view

Figure 3: Render of version 1 of the bench showing a dovetail interface.

1.5 Demonstrator

As a full size bench would be too large to reasonably fit in the manulab, a scaled demonstrator is proposed. It will be be made of 3D-printed components and/or laser cut wood sheets.

2 Manufacturing concept

2.1 Flow chart

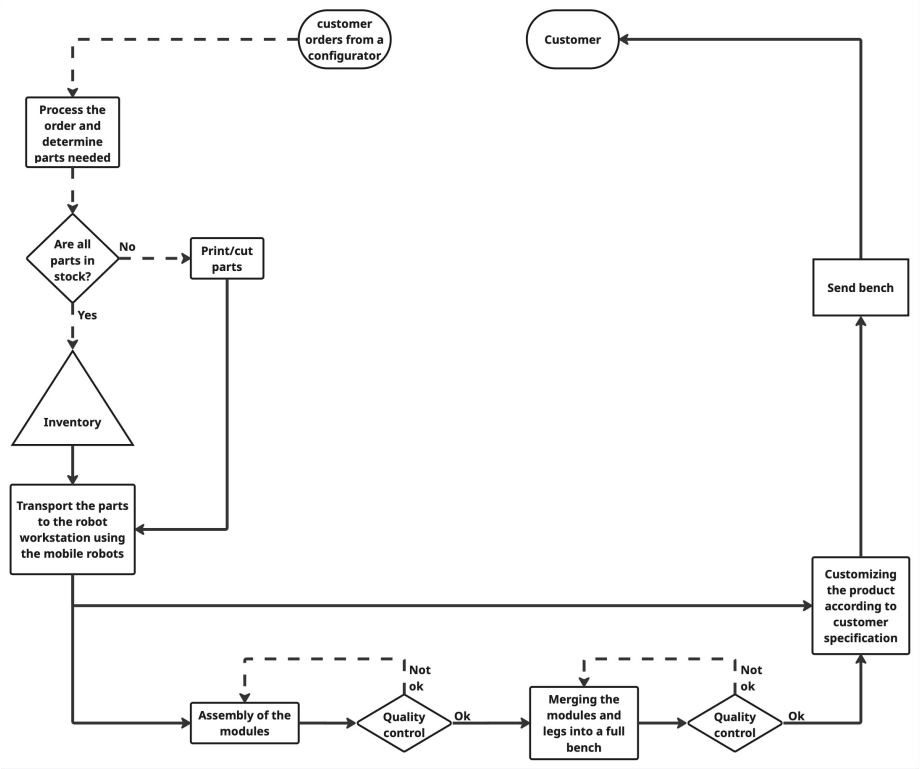


Figure 4: First draft of a flow chart for the current product concept (drawn in Miro).

Using a configurator on a website, the customer will be able to customize and order the furniture according to their specific needs. To fulfill the order, the parts needed for the products will be determined. If all the needed parts are in stock, a mobile robot will simply gather the components from the inventory. If they are not in stock, the part will be laser cut and or printed and then transported to the robot cell. Some customization to the product, like engraving or similar, can be added in the laser cutting and 3D-printing step. The robot(s) will then assemble the bench in several steps and perform quality controls in between the steps. The final customization is then added to the parts before shipping it to the customer.

As lean has not yet been covered in class, the Lean analysis has intentionally been omitted at this stage.

3 Preliminary project plan

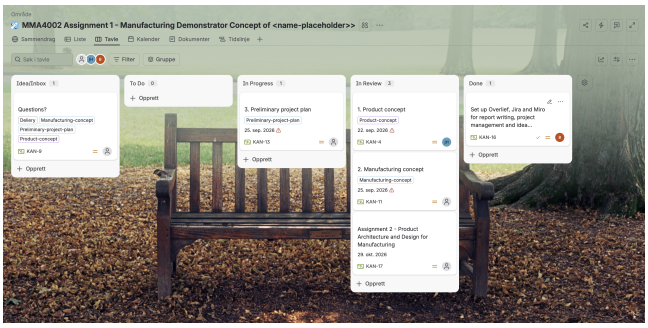


Figure 5: Screenshot from Jira.

Project planning is done using the planning tool Jira figure 5.

- The first step is to complete the cad files of the furniture and decide what mechanisms and common architecture to use. We are still behind on that because we started pretty late (only one week before delivering this report). Therefore, the ideas presented in the report are still very rough and not fully complete. This should ideally be finished no later than around the 10th of October.
- The next step would be to design the jigs while developing a manufacturing procedure that would work in the manulab. This should be more or less finalized by the end of October.
- The final step is to finalize and perform the procedure and film it. This should be done at least one week before the deadline of the project.

Bibliography

Mouratidis, Kostas (2026). "Aesthetic preferences for traditional versus contemporary architecture: Cross-national evidence from real-world buildings". In: *Cities* 179, p. 107410. ISSN: 0264-2751. DOI: https://doi.org/10.1016/j.cities.2026.107410. URL: https://www.sciencedirect.com/science/article/pii/S0264275126006426.

Appendix

A Website

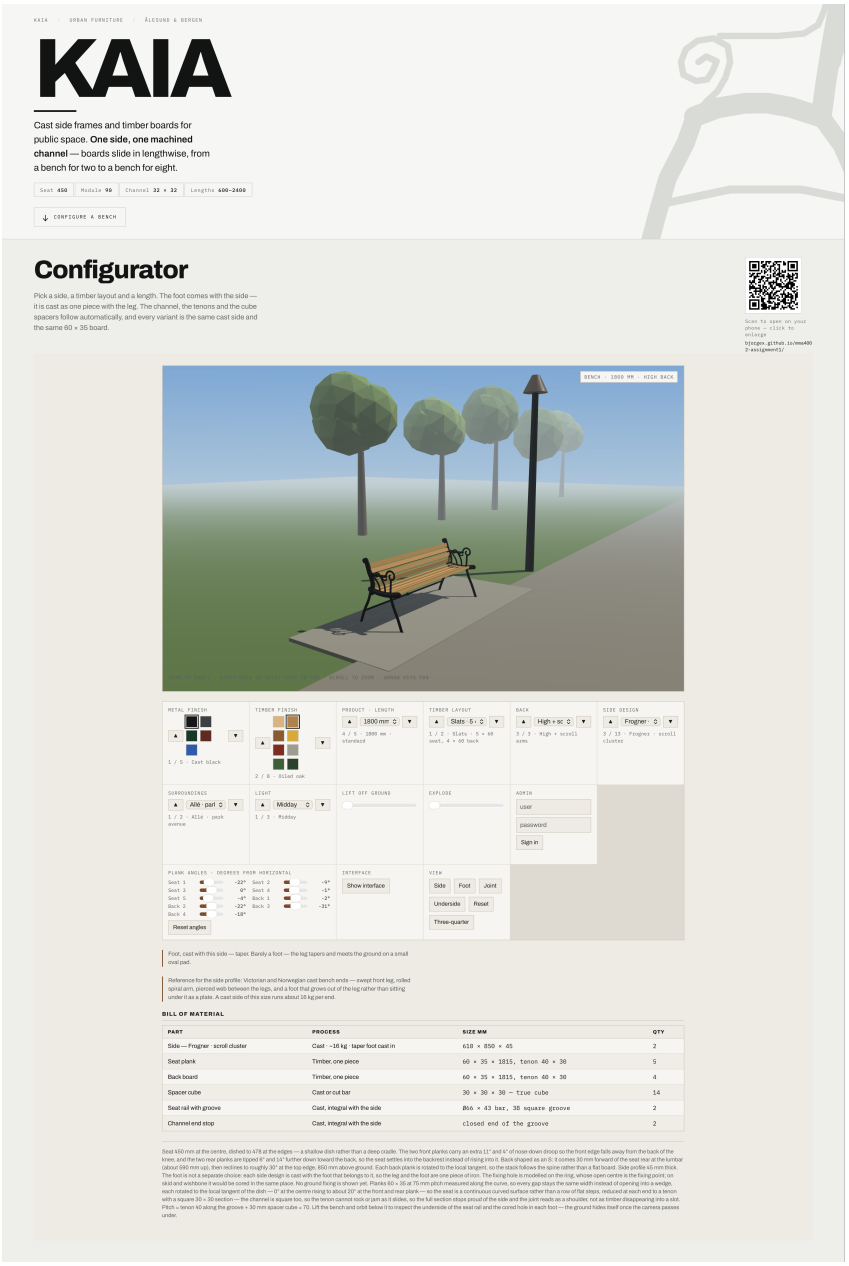


Figure 6: First draft of a configurator website. Developed using AI.